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The influence of a neotropical herbivore (*Lamponius portoricensis*) on nutrient cycling and soil processes

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Abstract The role of phytophagous insects in ecosystem nutrient cycling remains poorly understood. By altering the flow of litterfall nutrients from the canopy to the forest floor, herbivores may influence key ecosystem processes. We manipulated levels of herbivory in a lower montane tropical rainforest of Puerto Rico using the common herbivore, *Lamponius portoricensis* (Phasmatodea), on a prevalent understory plant, *Piper glabrescens* (Piperaceae), and measured the effects on nutrient input to the forest floor and on rates of litter decomposition. Four treatment levels of herbivory generated a full range of leaf area removal, from plants experiencing no herbivory to plants that were completely defoliated ($>4,000\text{ cm}^2$ leaf area removed during the 76-day study duration). A significant ($P < 0.05$) positive regression was found between all measures of herbivory (total leaf area removed, greenfall production, and frass-related inputs) and the concentration of NO_3^- in ion exchange resin bags located in the litter layer. No significant relationship was found between any of the herbivory components and resin bag concentrations of NH_4^+ or

PO_4^- . Rates of litter decay were significantly affected by frass-related herbivore inputs. A marginally significant negative relationship was also found between the litter mass remaining at 47 days and total leaf area removed. This study demonstrated a modest, but direct relationship between herbivory and both litter decomposition and NO_3^- transfer to the forest floor. These results suggest that insect herbivores can influence forest floor nutrient dynamics and thus merit further consideration in discussions on ecosystem nutrient dynamics.

Keywords Decomposition · Herbivory · Insects · *Piper glabrescens* · Puerto Rico

Introduction

Insect herbivores are a prevalent component of forest ecosystems and a key factor in regulating canopy processes (Schowalter et al. 1986; Huntly 1991; Rinker et al. 2001). Their high mobility and rapid rates of reproduction allow them to quickly respond to alterations in their environment and exploit emerging resources. Selective, and often dramatic, feeding by insect herbivores exerts a strong influence on a number of forest ecosystem properties (Schowalter 2000). In addition to the well-documented and conspicuous effects that they can have on plant growth and survival, herbivores can substantially alter plant species composition (Ritchie et al. 1998) and the flow of nutrients within forest systems (Swank et al. 1981; Fogal and Slansky 1985; Hollinger 1986; Schowalter et al. 1991; Hunter 2001).

Phytophagous insects directly modify nutrient flows by altering the timing, quantity and quality of canopy inputs to the forest floor. Herbivore-derived inputs are commonly in the form of insect feces (frass), green leaf fragments (greenfall) and modified throughfall, all of which form a chemically distinct component of canopy inputs to the forest floor. Insect frass has been shown to contain large fractions of labile C and N (Lovett and Ruesink 1995), whereas greenfall can have as much as

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twice the N and P found in senescent leaves (Grace 1986; Lodge et al. 1991; Risley and Crossley 1993), leading to more rapid rates of decomposition (Fonte and Schowalter 2004). The enrichment of canopy throughfall may be important as well. By damaging leaves and exposing inner leaf tissues to leaching, herbivores have been shown to increase throughfall concentrations of K, N, S, Ca and dissolved organic C (Kimmins 1972; Seastedt et al. 1983; Schowalter et al. 1991; Reynolds et al. 2000; Stadler et al. 2001). These herbivore-induced alterations generally increase the availability and often the quantity of nutrients that fall on the forest floor and could further modify ecosystem nutrient flow by altering patterns of nutrient input to the soil subsystem.

The effects of herbivore-derived inputs on soil processes and nutrient cycling are variable and poorly understood, but are likely to depend on the nature of the inputs and ecosystem conditions. Although a number of experiments have demonstrated modification of canopy inputs by insect herbivores (e.g., Kimmins 1972; Ohmart et al. 1983; Fogal and Slansky 1985; Grace 1986; Hollinger 1986; Risley and Crossley 1988; Lightfoot and Whitford 1990; Stadler and Michalzik 2000), few have addressed the effects of these modified inputs on the soil subsystem or plant growth (Seastedt et al. 1983; Schowalter et al. 1991; Lovett and Ruesink 1995; Belovsky and Slade 2000; Reynolds and Hunter 2001; Chapman et al. 2003; Hunter et al. 2003). Some have speculated that the addition of herbivore inputs could accelerate rates of nutrient cycling and even enhance primary productivity (Mattson and Addy 1975; Kitchell et al. 1979). However, experimental field studies (Seastedt et al. 1983; Schowalter et al. 1991; Reynolds and Hunter 2001) have produced inconclusive results and none has involved tropical ecosystems.

In this study, we manipulated levels of herbivory in a tropical forest using walkingsticks (*Lamponius portoricensis*, Phasmatidae) as a representative herbivore. We evaluated the effects of leaf area loss on rates of litter decomposition and nutrient fluxes to the forest floor. We expected elevated levels of herbivory to increase the flow of available nutrients (N and P) to the forest floor and accelerate the rates of litter decomposition.

Materials and methods

Study site

This study was conducted at El Verde Field Station, Luquillo Long-Term Ecological Research (LTER) Site in Puerto Rico (18°10'N, 65°30'W). The site is classified as a subtropical wet forest, with average temperatures ranging between 21°C in January and 25°C in August (Waide and Reagan 1996). Annual precipitation averages 3,700 mm and varies seasonally, with 200–250 mm month⁻¹ January–April (dry season) and 350 mm month⁻¹ the remainder of the year (McDowell and Estrada-Pinto 1988). Precipitation exceeds evapo-

transpiration in all months. Soils in this region are dominated by Ultisols belonging to the Los Guineos series, which are well-drained, acidic clays and silty clay loams (Waide and Reagan 1996). The study site was located at approximately 300 m elevation in a tabonuco (*Dacryodes excelsa*, Burseraceae) forest type dominated by *D. excelsa*, *Manilkara bidentata* (Sapotaceae), *Sloanea berteriana* (Elaeocarpaceae), *Guarea guidonia* (Meliaceae) and *Prestoea montana* (Arecaceae) in the upper canopy. Detailed description of the vegetation is available in Lawrence (1996), Odum and Pigeon (1970) and Waide and Reagan (1996).

The study site was characterized by relatively high abundance of *Piper glabrescens* (Piperaceae), a common understory plant and a primary food source of the walkingstick, *L. portoricensis* (Willig et al. 1993). This plant–herbivore interaction was chosen for study because both species are common, consistently associated, easy to manipulate, and relatively resilient to minor disturbances.

Treatments

Forty *P. glabrescens* individuals of similar size, apparent condition (based on foliage color and absence of obvious injury or disease), and foliage area were selected within a 0.5 ha area. Individual shrubs were distinct and generally separated by 1–3 m. Each plant was roughly 1 m in height and possessed a median value of 16 leaves (range 10–33; plants with fewer leaves had larger leaves). Cubic enclosures (60×60×60 cm³, 3.2 mm nylon mesh) were placed around the foliage of each *P. glabrescens* and suspended above the ground from a 2-m PVC pole. Each enclosure was carefully fitted, allowing room for plant growth, and sealed to restrict the passage of large herbivores (mainly *L. portoricensis* adults).

The plants were randomly assigned to one of four treatments (ten replicates each): high and low levels of herbivory, herbivore exclusion, and control. The controls had large (25×25 cm²) openings cut in the underside of the cage to allow free access to herbivores, while the upper portion and sides were left intact in order to maintain microclimatic and shading conditions similar to the other treatments. The low-herbivory treatment contained one adult male *L. portoricensis* in each cage, and the high-herbivory treatment had one adult male and one adult female *L. portoricensis* in each cage. Both sexes were used in the high herbivory treatment due to the difficulty in collecting sufficient numbers of adult specimens of the same sex. The resulting densities of walkingsticks on treated plants were similar to densities of 0.4–2.9 walkingsticks m⁻² reported by Willig et al. (1986) and Willig and Camilo (1991) at this site prior to Hurricane Hugo. Prior feeding experiments with *L. portoricensis* in captivity (S. J. Fonte, unpublished data, Sandlin and Willig 1993) indicated that these treatment levels would yield adequate levels of herbivory. All enclosures were visited every 2–4 days in order

to maintain the prescribed herbivore levels and to remove litter from the tops of the cages.

Herbivory measurements

The total leaf area and leaf area missing for five randomly chosen leaves of each plant were measured three times: prior to treatment, after 1 month, and after 2 months. A non-destructive point intercept method was used to measure leaf area. Leaves were placed under a transparent sheet with a printed 5×5 mm grid. The number of points intersecting the leaf and areas missing from the leaf were then tallied, with each point representing the center of a 5×5 mm² square. The selected leaves were tagged and measured again at each sampling time, unless they were missing, in which case a new leaf was assigned. The mean leaf areas and areas missing for the five leaves were multiplied by the total number of leaves on a plant in order to obtain an estimate of the total leaf area and the leaf area missing for each plant. A quick visual inspection confirmed that herbivore damage on these leaves was representative of the entire plant. Total herbivory was calculated by adding the area of leaves missing entirely to the area lost on the remaining leaves. Emerging leaves (including leaves with an area of less than 2 cm²) were monitored, but not included in the total leaf area estimates until they had expanded and matured.

Total herbivory, expressed as square centimeters of leaf area removed, was subdivided into two components of herbivore inputs, greenfall (consisting of green leaf fragments larger than 5×5 mm²) and frass-related inputs (calculated as the difference between total leaf area removed and the leaf area removed as greenfall). The portion of the leaf area removed in the form of greenfall was calculated by collecting fallen leaf fragments trapped in the enclosures every 2–4 days. Due to rapid dehydration and shrinkage of this material, the greenfall area could not be measured directly using the point intercept method. For each green leaf fragment, we estimated the area as a proportion of a whole leaf, multiplied this by the average area of a leaf for each plant, and added the area of all fragments in order to provide an estimate of total greenfall area per plant. Although the area of individual leaves differed greatly between plants, the leaf size was relatively uniform among the leaves on any particular plant, thus allowing for this quick and relatively accurate estimate of area. Following this measurement, greenfall material was scattered beneath the corresponding plant.

The greenfall component calculated in this manner is likely to underestimate true greenfall deposition, due to our inability to detect and measure the smallest fragments that fell through the mesh. In addition, greenfall estimates for the control treatments may be artificially low due to the potential for material to fall through the openings on the underside of the enclosures and escape measurement. Likewise, the deposition of frass-related

inputs (the non-greenfall component) overestimates actual quantities of frass, since this number includes small leaf fragments and plant material assimilated by the walkingsticks. However, visual observation suggested that most of this component was indeed comprised of frass. It should also be noted that our calculation of the frass component as the difference between total leaf area removed and greenfall is subject to compounded variation and should be interpreted with some caution.

Ion-exchange resin bags

Ion-exchange (IE) resin bags provide a useful tool for measurement of available soil nutrients (Binkley and Matson 1983; Binkley et al. 1986). They function by adsorbing dissolved nutrient ions onto charged resin beads as the solution passes through the bags (Skogley and Dobermann 1996). The resin bags used in this study were fabricated from nylon stockings and contained one compartment each of positively and negatively charged resin beads. Each compartment was filled with 10 g of resin and covered an area of approximately 16 cm². Due to the high variability of substrate (litter, rock, roots, woody debris) below each *P. glabrescens*, the resin bags could not be buried uniformly in the traditional manner. Instead, each resin bag was placed on the forest floor surface, centered directly under each individual *P. glabrescens* canopy, in early March 2002. Although not providing a direct measure of soil nutrient availability, resin bags placed in this manner provided relative comparisons of nutrient inputs to the soil from above. They recorded nutrients dissolved in throughfall or released, via leaching or mineralization, from greenfall and frass deposited on top of the bags during the course of the experiment.

Resin bags were collected after 64 days. Debris was removed before placing them into individual plastic bags. The bags were placed on ice and shipped overnight for analysis. Nutrient ions (NO₃⁻, NH₄⁺ and PO₄⁻) were extracted from each resin bag in 100 ml of 1 M KCl and analyzed using automated cadmium reduction, phenate, and automated ascorbic acid reduction methods, respectively (Greenberg et al. 1992) on an Alpkem Flow-Injection analyzer (ANKOM, Fairport, NY, USA).

Litterbag preparation

Although litterbags may provide artificial estimates of leaf decomposition rates due to the partial exclusion of soil fauna, the air-drying of leaves, and alterations to litter microclimates, they are a useful tool for comparing relative rates of decomposition between treatments (Swift et al. 1979; Taylor 1998; González and Seastedt 2001). Recently senesced leaflets of *D. excelsa* were collected from the forest floor several weeks prior to the start of the experiment. This species was used because it is prevalent in the forest and its litter is easily collected in

sufficient quantities over a limited area within a minimal time. Immediately following the collection, leaflets were air-dried for 48 h in a room with a dehumidifier and thoroughly mixed before weighing. Litterbags (10×10 cm, 1 mm nylon mesh) were filled with 2.5 g $\pm$ 0.1 g air-dried litter. Thirty leaf samples were collected at regular intervals during litterbag construction in order to obtain an oven-dry (50°C) weight conversion for the air-dried leaves.

At the start of the experiment, in early March 2002, four litterbags were placed beneath each *P. glabrescens* individual (around the resin bags) and picked up after 9, 18, 47 and 76 days in the field. All litterbags were oven-dried at 50°C for 7 days upon collection. The litter was carefully separated from roots, soil and other debris and weighed in order to determine the mass remaining.

Data analysis

Simple linear regression was used to generate first-order decay rate constants, k , on a per year basis (Olson 1963) separately for the groups of four litterbags under each plant. A natural log transformation of the herbivory components (total leaf area removed, greenfall and frass-related inputs) best met the assumptions for regression analysis against k (Ramsey and Schafer 1997). Treatment differences for total herbivory and the various herbivore-input components were analyzed using one-way ANOVA. ANOVA was also used to investigate treatment differences in decay rates (k) and resin-bag ion concentrations (NO_3^- , PO_4^- and NH_4^+). Simple linear regression was employed to evaluate relationships between decay rates and resin-bag ion concentrations and the various herbivory input components. All analyses were performed using S-Plus statistical software (MathSoft 2000).

Results

Herbivory

Treatments produced significantly different levels of leaf area removal ($F=37.5$, $df=3,36$, $P<0.001$). As antici-

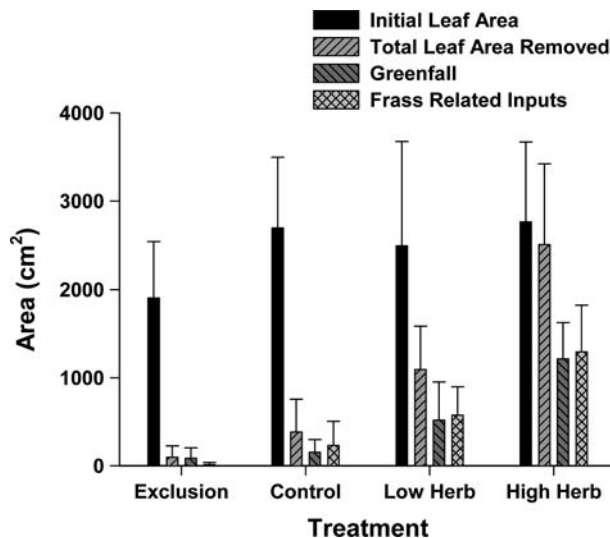


Fig. 1 Comparison of herbivory components on *Piper glabrescens* among herbivore (*Lamponius portoricensis*) treatments during a 76-day experiment in a lower montane tropical forest in Puerto Rico in 2002. Error bars denote the SD ($n=10$)

pated, total leaf area removed was lowest in the herbivory exclusion treatment ($<10\%$ of total) and greatest in the high herbivory treatment, with many plants experiencing complete foliage removal. However, large variation within each treatment resulted in a continuum of herbivore inputs (leaf area removed) across all treatments rather than discrete levels (Fig. 1). This probably reflected differences in plant biochemical condition, or other unmeasured sources of variation, and required analyses relying primarily on simple linear regression.

Walkingsticks in the elevated herbivory treatments removed an average of $22.1 \text{ cm}^2 \text{ g}^{-1}$ live insects day^{-1} during the course of the study. Greenfall was largely in the form of nearly intact leaves that had been chewed off near the petiole (the area of greenfall fragments averaged 81% of an intact leaf). Greenfall accounted for 46% of the total leaf area lost (Table 1). Potential differences in feeding rates between the high and low herbivory treatments were anticipated, because walkingsticks in the high herbivory treatments may have experienced resource limitation, territoriality and sexual behavior to a greater extent than did walkingsticks in the

Table 1 Average consumption rates and feeding behavior of the walkingstick, *Lamponius portoricensis* feeding on *Piper glabrescens*, during a 76-day experiment in a lower montane tropical rainforest in Puerto Rico during 2002. The SD is shown below each value in italics

Total leaf area removed ($\text{cm}^2 \text{ g}^{-1} \text{ day}^{-1}$) ^a			Greenfall ($\text{cm}^2 \text{ g}^{-1} \text{ day}^{-1}$) ^a			Frass-related inputs ($\text{cm}^2 \text{ g}^{-1} \text{ day}^{-1}$) ^a		
Low ^b	High ^c	Avg ^d	Low ^b	High ^c	Avg ^d	Low ^b	High ^c	Avg ^d
24.0	20.3	22.1	10.5	10.0	10.3	13.4	10.3	11.9
<i>11.1</i>	<i>8.3</i>	<i>9.8</i>	<i>7.7</i>	<i>4.3</i>	<i>6.1</i>	<i>10.0</i>	<i>4.3</i>	<i>7.7</i>

^aUnits are cm^2 of leaf area removed per g of live *L. portoricensis* per day

^bLow herbivory treatment = one adult male *L. portoricensis* ($n=10$)

^cHigh herbivory treatment = one adult male and one adult female *L. portoricensis* ($n=10$)

^dAvg = average values from high and low treatments combined ($n=20$)

low herbivory treatment. However, differences in total herbivore inputs, greenfall, and frass-related inputs ($\text{cm}^2 \text{g}^{-1} \text{live insect day}^{-1}$) between the two treatments were not significant ($P > 0.10$).

Nutrient flux

The resin bags revealed that N inputs in the form of NH_4^+ to the forest floor greatly exceeded N inputs as NO_3^- and P in the form of PO_4^- (Fig. 2). Although the high herbivory treatment yielded the highest average concentrations of both NO_3^- and NH_4^+ in resin bags, comparisons in ANOVA revealed no significant differences among treatments for any of the nutrient ions. Nitrate was the only ion that was significantly related to herbivore inputs. The total leaf area removed, greenfall, and frass-related inputs estimated at 64 days were significantly related to NO_3^- levels ($F = 6.62$, $\text{df} = 1,37$, $P = 0.014$, $R^2 = 0.15$; $F = 5.53$, $\text{df} = 1,37$, $P = 0.024$, $R^2 = 0.13$; $F = 6.24$, $\text{df} = 1,37$, $P = 0.017$, $R^2 = 0.14$ respectively). The relationship between leaf area removed (cm^2) and nitrate concentration in the resin bags (mg l^{-1}) was described by the equation:

$$\text{NO}_3^- = 0.0002 \times \text{Total leaf area removed} + 0.798$$

indicating that each $1,000 \text{ cm}^2$ increase in leaf area loss resulted in a 0.2 mg l^{-1} increase in NO_3^- concentration in resin bags (Fig. 3). None of the herbivory measures were significantly related to NH_4^+ or PO_4^- ($P > 0.10$).

Decomposition rates

With an average mass loss of over 35% after 76 days, decay in this experiment proceeded rapidly. The regression lines used to generate decay rate constants, k , were

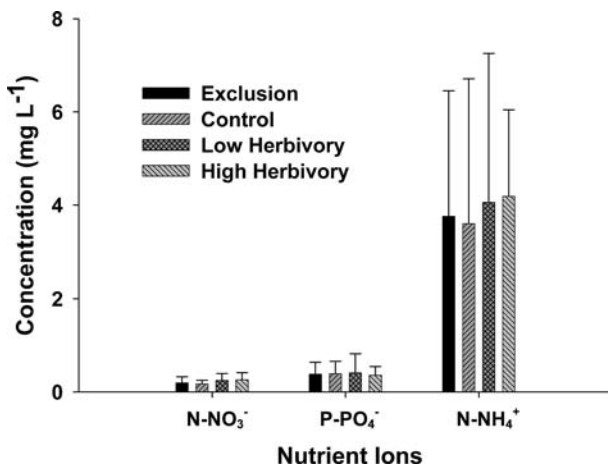


Fig. 2 Mean nitrogen and phosphorus concentrations in ion exchange resin bags placed on the forest floor under four treatments of varying herbivory after 64 days in a lower montane tropical rainforest in Puerto Rico. Error bars denote the SD ($n = 10$)

highly significant ($P < 0.01$) across all treatments (Fig. 4), except for three cases where missing litterbag weights yielded exaggerated k values, prompting these study units to be dropped from the analysis. Although the regressions for mass loss by herbivory treatment were not significant, the decay constants increased from exclusion to high herbivory treatments, as expected (Fig. 4).

Neither $\ln(\text{total leaf area removed})$ nor $\ln(\text{greenfall deposited})$ explained the rates of decay measured in the litterbags ($P > 0.14$). However, the natural log of frass-related inputs, defined as $\ln(\text{total leaf area removed} - \text{greenfall})$, was significantly related to decay rates ($F = 5.53$, $\text{df} = 1,35$, $P = 0.034$, $R^2 = 0.12$). This relationship is described by the equation:

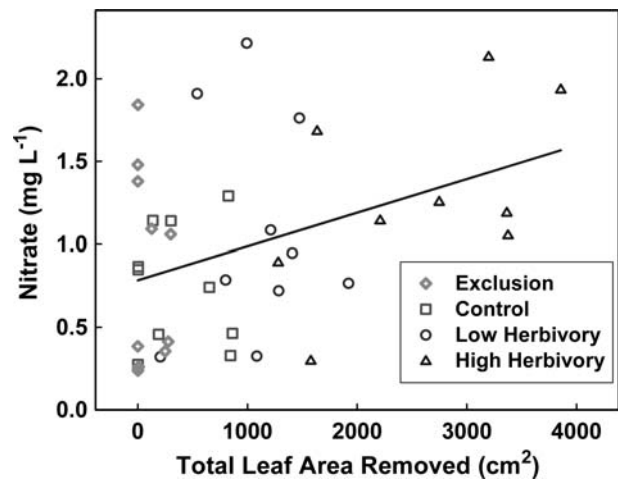


Fig. 3 Concentration of nitrate (NO_3^-) in ion exchange resin bags vs. the total leaf area removed by *Lamponius portoricensis* feeding on *Piper glabrescens* after 64 days in a lower montane rainforest in Puerto Rico in 2002

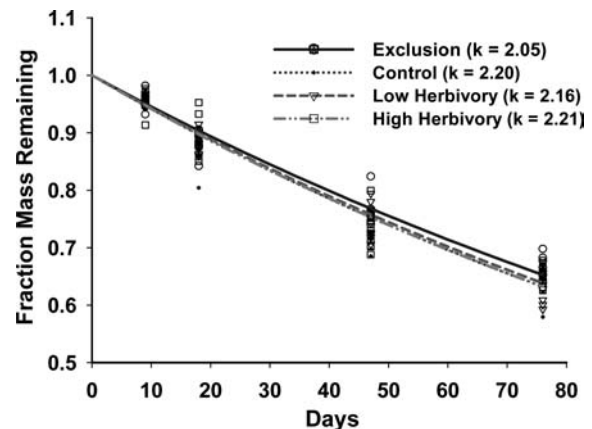


Fig. 4 Mass loss of *Dacryodes excelsa* foliage litter in litterbags on the forest floor under *Piper glabrescens* subjected to four herbivory treatments, with *Lamponius portoricensis*, during a 76-day experiment in a lower montane rainforest in Puerto Rico in 2002. Lines represent regression models

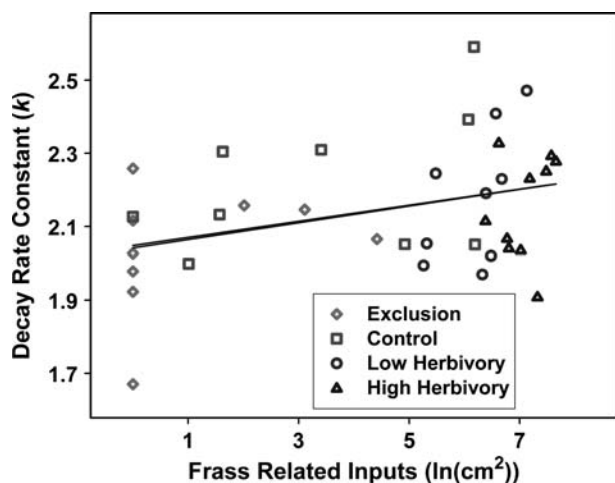


Fig. 5 Decay rate constant (k) vs. the natural log of frass-related inputs (total leaf area removed—leaf area removed in the form of greenfall) by *Lamponius portoricensis* feeding on *Piper glabrescens* after 64 days in a lower montane rainforest in Puerto Rico in 2002

$$k = 0.022 \times \ln(\text{Frass related inputs}) + 2.048$$

implying that for every doubling of frass-related inputs the mean of the decay rate constant increases by 0.015, or 0.75% (Fig. 5). Treatment comparisons using ANOVA failed to explain variability in k ($P=0.17$).

Due to the complete defoliation of some plants before the end of the experiment, herbivore-generated inputs did not fall continuously throughout the entire study for all of the plants in the high herbivory treatment. The retrieval of the third set of litterbags, after 47 days, coincided with the first plant becoming completely defoliated. Therefore, we examined the effects of herbivore inputs on mass loss at 47 days, before inputs from some of the higher herbivory treatments were discontinued. A significant negative relationship between $\ln(\text{frass inputs})$ and the litter mass remaining at 47 days ($df=1,38$, $P=0.044$, $R^2=0.10$) is consistent with results reported above using a single k value generated from four litterbags under each plant. A marginally significant negative relationship was also found between $\ln(\text{total leaf area removed})$ and litter mass at 47 days ($df=1,38$, $P=0.069$, $R^2=0.08$). Despite evidence of herbivore effects on nutrient flux and decay rate, we found no significant ($P>0.10$) relationships between decomposition and nutrient ions in resin bags.

Discussion

The results from our study suggest that herbivores can influence nutrient cycling and key soil processes in a tropical forest ecosystem. All three measures of herbivory in this study were significantly related to NO_3^- concentrations in resin bags. Also, there was a modest, but significant, relationship between frass-related inputs (total leaf area removed – greenfall) and decomposition rates. The low amount of variability explained by our

data suggests that herbivory may not play a driving role in ecosystem nutrient dynamics at the levels of herbivore inputs that we tested. These relationships, however, are statistically significant and indicate the potential for herbivore inputs to have important ecosystem consequences.

Relative to other factors that regulate nutrient availability and decomposition within this forest, herbivore inputs appear to play only a secondary role. Given the large degree of spatial heterogeneity present at our site, considerable noise (as well as low R^2 values) was expected from our data. The species composition and structure of the overstory column centered on any given plant differed greatly among plants. The soil surface at the study site was extremely variable, with patches of roots, rock, woody debris, exposed soil and leaf litter. In addition to overstory composition and substrate variability, intensive precipitation events occur with high frequency in this forest, and the volume of water flowing across any particular surface can vary greatly, depending on canopy architecture and microtopography. Variation in biochemical interactions between plants and walkingsticks are likely to have contributed to variability in herbivore-mediated inputs. The numerous sources of background variability present in this study may have masked herbivore-induced effects and could account for the low amount of variability explained by herbivore-derived inputs. The significant trends that do emerge from this variability therefore merit closer examination. The fact that consistent trends arose among multiple response variables suggests that these effects are ecologically relevant.

Although the magnitude of influence exerted by herbivore inputs on soil processes seems small in this study, we argue that the potential implications are important. Under higher levels of herbivore inputs we suspect that the effects would become more pronounced. The complete defoliation of a single *P. glabrescens* in the understory (as occurred in this study) represents only a fraction of the total leaf area contained in a vertical column of tropical forest canopy (Lawrence 1996; Richards 1996). Hence, even low or moderate levels of herbivory distributed through the vertical column of canopy may supply quantities of herbivore-generated inputs equal to or exceeding those tested in this study. Herbivore outbreak situations offer the most relevant circumstance for this argument. Although insect outbreaks have not been recorded as often in tropical forests as in temperate forests, Van Bael et al. (2004) reported outbreaks may be common in tropical forests, but are more pronounced, short-term events. Torres (1992) and Schowalter (personal observations) noted outbreaks of several lepidopteran species on understory vines and forbs at Luquillo following Hurricanes Hugo and Georges, respectively. In addition, we suggest that even small aberrations in key processes such as nutrient transfer and litter decomposition can have significant ecosystem impacts when considered over long time intervals. These long-term implications could include

alterations to soil carbon storage, soil fertility or plant community succession.

A number of studies have suggested that canopy herbivore inputs can influence a range of soil properties. For example, Hunter et al. (2003) found positive relationships between background levels of frass deposition and N availability in the soil of a temperate deciduous forest. In the same forest, Reynolds and Hunter (2001) manipulated background levels of throughfall, greenfall, and frass and found these inputs to differentially influence soil microbial activity (respiration) and nutrient availability, as well as nematode and collembolan populations in the litter layer (Reynolds et al. 2003). Schowalter and Sabin (1991) found that herbivory altered litter microarthropod communities in a coniferous forest ecosystem. Such alterations to soil fauna communities, soil microbial activity or nutrient availability can influence key processes such as nutrient uptake by plants (Marschner 1995), organic matter turnover (Paul and Clark 1996) or ecosystem nutrient loss (Swank et al. 1981; Lovett et al. 2002). Thus, our results corroborate previous findings and indicate that herbivores can influence a number of soil and ecosystem attributes.

The most pronounced effect of herbivory in this study was on the transfer of nitrate to the litter layer. However, other nutrient forms are likely to be influenced by herbivory as well (Kimmins 1972; Hollinger 1986). Background concentrations of NH_4^+ and total P in throughfall are higher and more variable than that of NO_3^- in this system (McDowell 1998) and perhaps masked any treatment effects on NH_4^+ and PO_4^- . Throughfall from the upper canopy is likely to have contributed to the high variability and low R^2 values observed for NO_3^- as well.

It is not surprising that frass-related inputs were the only component of herbivory that were significantly related to decomposition. Comprised primarily of insect frass, and to a lesser extent small greenfall fragments, this material should release nutrients more rapidly than would large greenfall material or senescent leaf litter due to a much greater surface area to volume ratio and differing chemical compositions. Several studies have suggested that frass may have different effects than does leaf litter upon deposition to the soil surface. Gypsy moth frass was found to contain large fractions of labile C and N and to stimulate microbial activity (respiration) when added to soil (Lovett and Ruesink 1995). Christenson et al. (2002) showed that N in frass was rapidly distributed throughout multiple soil and plant-N pools upon deposition, whereas leaf litter-N was largely retained in the decaying leaves and released only slowly. In plant-soil microcosms, Frost and Hunter (2004) found frass additions to increase total soil C, total N, NH_4^+ in the soil and NO_3^- in leachate. We also would expect frass-related inputs to be highly related to leaching losses from chewed leaves remaining on the plant, since the production of frass and small greenfall fragments is associated with more intensive leaf damage and subsequent exposure of inner leaf tissues. Greenfall, as

measured in this study, would probably play a less active role in short-term litter nutrient dynamics, due to its large size and intact nature.

The relationship between NO_3^- levels and herbivore inputs suggests that increased availability of N may have played a role in accelerating litter decay rates (Schlesinger and Hasey 1981; Hunt et al. 1988; Enriquez et al. 1993). However, decay rates were not related to NO_3^- concentrations, indicating that other factors, such as organic N, may regulate decomposition processes. Both Mg and Ca have been recognized as important predictors of decay at El Verde (Bloomfield 1993; Bloomfield et al. 1993). Given that Ca flow from the canopy has been linked with herbivory in other studies (Schowalter et al. 1991), other nutrients may be more important than originally considered. Decay processes are also affected by alterations to soil communities (Coleman and Crossley 1996; Wardle and Lavelle 1997; Heneghan et al. 1999). Hence, the accelerated decay associated with herbivore inputs in this study may be the net result of multiple factors. Decomposition in our study was comparable to the decay of *D. excelsa* reported from other studies (Sullivan et al. 1999; Fonte and Schowalter 2004).

In conclusion, this study presents evidence for a direct link between herbivory and nutrient cycling in a tropical forest ecosystem. Herbivory on a small understory shrub had modest, but significant, effects on both NO_3^- flux and decomposition rates. Although the observed effects were small, they indicate that herbivore inputs could play a more prominent role under more intensive levels of herbivory or when observed at longer time scales. Because of their potential to influence ecosystem nutrient dynamics, canopy herbivores merit further consideration in the management and general understanding of tropical forest ecosystems.

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